Find the signal. Keep the auditor in control.

FINTECH AI — Digital Surfers | Evidence-led investigation of unusual financial transactions

THE PROBLEM

- High-volume transaction spreadsheets make unusual items hard to spot.
- A raw anomaly flag lacks context and follow-up evidence.
- Review decisions and investigation history need a consistent record.

INTENDED USERS

- Internal / external auditors
- Finance control and compliance teams
- Audit leads who need a review trail

AI assists investigation; it does not decide fraud or replace professional judgment.

PROPOSED SOLUTION

Import → screen → explain → review

Deterministic data checks work without AI. Gemma is an optional evidence-grounded explanation layer.

01 IMPORT

Validate CSV rows and persist transactions.

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02 SCREEN

Flag repeated references and supplier + amount + date matches.

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03 EXPLAIN

Gemma interprets server-supplied evidence; no database/tool access.

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04 REVIEW

Auditor confirms or dismisses; decision and timestamp are stored.

EVIDENCE BOUNDARY

Observed evidence is persisted transaction / rule output. Gemma contributes interpretation and recommended checks—not new transaction facts or financial actions.

One audit API. Cloud Gemma + local Gemma.

Next.js / React → FastAPI / Pydantic → SQLite. AI adds interpretation; deterministic screening remains independent.

FRONTEND

- Dashboard + CSV upload
- Transactions + findings
- Same-origin API proxy

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API + DATABASE

- FastAPI / HTTPX / SQLite
- Transactions + findings
- Explanations + reviews

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CLOUD: LIVE

- Google AI Studio
- gemma-4-26b-a4b-it
- App Explain uses cloud

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LOCAL: LIVE

- Ollama: gemma4:latest
- 7.5B Q4_K_M / 6.6GB
- Remote /ollama API access

REPRODUCIBLE EVIDENCE

3 backend tests passed • public cloud explain + local generation succeeded • separate Pandas / Isolation Forest sample: 100 accepted, 6 duplicate rows, 34 anomaly rows (not yet wired into upload).

LIVE DEMO / 90 SECONDS

CSV → duplicate evidence → Gemma explanation → review

Public app: ranks-atlanta-formerly-diverse.trycloudflare.com/dashboard

SHOW THE END-TO-END FLOW

- Upload 01_mixed_office_transactions.csv (20 valid rows).
- Open Anomalies & AI; inspect a duplicate candidate.
- Explain: Google-hosted Gemma interprets the supplied evidence.
- Save review; refresh to show persisted status.
- Local model: POST /ollama/api/generate; Swagger: /backend/docs

THREE SYNTHETIC DEMO FILES

- 01 Office: 20 valid rows; varied costs + duplicate candidates.
- 02 Retail: 20 valid rows; inventory, logistics + duplicates.
- 03 Projects: 16 valid + 4 invalid rows; demonstrate rejection reporting.
- Large payment ≠ fraud. Same-value invoices may be legitimate; auditor checks.

CSV downloads: /demo-data/ on the same host. Public smoke test passed upload → finding → saved review; local and cloud generation both succeeded.

Faster triage is the goal. Traceable review is the deliverable.

Value: bring records, candidate findings, AI interpretation and recorded human decisions into one workflow.

DELIVERED & REPRODUCIBLE

- github.com/kwisdomk/FintechClankerz — public source; main + mbr-3 synced for demo.
- Frontend, FastAPI, SQLite, analysis pipeline, tests, reports + synthetic CSVs.
- Cloud + local Gemma responses verified; upload and review persist without AI.
- Measure next: triage time and review completeness versus spreadsheets; no savings claim yet.

LIMITATIONS & NEXT DEPLOYMENT

- Wire the full Isolation Forest pipeline into the upload API; current live screening is duplicate rules.
- Add team-approved OSS license, access controls, rate limits and provider data policy.
- DigitalOcean GPU + private Ollama: researched, NOT deployed. RTX 4000 Ada 20GB is a candidate to benchmark.
- Demo is temporarily unauthenticated; synthetic records only. Cloud Explain sends evidence to Google.

Cloud hosting today: Google AI Studio. Local inference today: developer machine + tunnel. DO research: digitalocean.com/products/gpu-droplets. Stop public access after the demo.